

## HPC Resources – Why HPC?

Utilizing HPC in your research can greatly accelerate it. Not only do HPC systems allow you to work with larger amounts of data, but they can also accelerate the process of making sense of it by providing more powerful computers. HPC systems expand the limits of what is possible with your research by allowing for more detailed simulations, more accurate models, and lightning-fast compute times. Instead of shrinking models to suit desktop workstations, HPC systems allow you to submit vast amounts of training data to a more capable machine. Problems that could take weeks can often be reduced to hours, enabling tighter feedback loops, faster iteration, and more ambitious experimental design.

HPC systems provide access to thousands of CPU cores, GPU accelerators, high-memory nodes, and ultra-fast interconnects designed specifically for large-scale computation. This means you can parallelize simulations, distribute workloads efficiently, and run ensemble experiments. Rather than settling for a single model run, you can execute hundreds of variations to test robustness. Instead of subsampling your dataset, you can analyze it in full. That shift, from “what can my laptop handle?” to “what does the science require?”, is extremely transformative in research—probably why [every Department of Energy lab to date utilizes them!](#)

## What is HPC?

**High-Performance Computing (HPC) systems** are powerful, interconnected computing systems. Instead of relying on a single processor, HPC systems combine thousands of CPUs and often GPUs across many nodes, connected by high-speed networks, to perform calculations in parallel. HPC environments typically include specialized hardware, large shared memory resources, high-performance storage systems, and job schedulers (such as [Slurm](#)) that efficiently distribute tasks across the cluster.

While a standard desktop machine might only have one CPU, HPC systems’ multi-core parallelism allows them to solve much more complicated problems than standard desktop machines.

**Some HPC systems are so large they take up [entire rooms!](#)**

## Benefits to Your Research

The primary benefit to using HPC systems to run computing tasks are:

- Faster processing speeds
- Larger data input handling
- Higher-resolution modeling
- Larger memory capacity
- Improved reproducibility through standardized software environments
- Allows access to scientific software stacks (ex: MPI, OpenMP, Cuda, etc.)

## Open-Source Systems

**Finding funding for HPC resources can be a struggle.** But if you are an academic researcher, there are many systems you can access through your .edu e-mail account **for free**.

Many HPC centers dedicated to scientific research provide free compute time on their systems through **allocation-based systems** where researchers can apply for storage space and compute time.

HPC systems can be ranked from small to incredibly large. Although even small, departmental-based clusters are typically enough for independent research tasks, HPC systems can typically be ranked in power as follows:

| Tier                          | Example                                       | Comparison                                    |
|-------------------------------|-----------------------------------------------|-----------------------------------------------|
| Departmental Cluster          | Small lab HPC                                 | Smaller<br>(100-2,000<br>cores)               |
| University Production Cluster | LEAP2 (Texas<br>State), AGES<br>(Texas A&M)   | Strong<br>(2,000-20,000<br>cores)             |
| National Academic HPC         | Stampede3<br>(TACC),<br>Perlmutter<br>(NERSC) | Larger<br>(50,000-300,000<br>cores)           |
| Leadership-Class              | Frontera / Frontier<br>(TACC)                 | Much larger<br>(300,000-millions of<br>cores) |

## Jargon

HPC systems often use jargon to describe their systems. Here are some common terms used:

- **Allocation** - The amount of computing resources formally granted to a researcher or research project for use on a shared supercomputing system.
- **Cluster** - A group of computers connected together so they function as one powerful system.

- **Node** - A single computer inside a cluster. Large HPC systems contain hundreds or thousands of nodes.
- **Compute Node** - A node dedicated to running calculations or research jobs.
- **Login Node** - A front-end machine where users prepare jobs, edit files, and submit work (not meant for heavy computation).
- **Core** - An individual processing unit inside a CPU capable of executing tasks independently.
- **GPU (Graphics Processing Unit)** - A specialized processor that performs many calculations simultaneously, commonly used for AI, simulations, and accelerated science.
- **Parallel Computing** - Running many parts of a problem simultaneously across multiple processors.
- **MPI (Message Passing Interface)** - A programming method allowing many nodes to communicate during large distributed computations.
- **Job** - A computational task submitted to the cluster.
- **Job Scheduler** - Software that decides when and where jobs run on the system (e.g., Slurm).
- **Queue (Partition)** - A waiting line for jobs grouped by resource type or priority.
- **Batch Job** - A job submitted to run automatically without user interaction.
- **Interactive Job** - A temporary session allowing direct interaction with compute resources.

## Local

### Texas State Cluster: LEAP2

Texas State University has [its own HPC cluster](#) available to Texas State researchers called LEAP2. With 5,808 total CPU cores, 8 NVIDIA A100 GPU nodes, and 100 GB/s, it is a respectable HPC system capable of rigorous scientific workloads. Any student or faculty member can request allocation space through the LEAP2 portal [here](#), though it is also open to requests for external collaborators.

Documentation for using the LEAP2 cluster can be found [here](#). It utilizes a SLURM scheduler and is Linux-based.

### Texas Advanced Computing Center (TACC)

If you are looking for more powerful systems, the Texas Advanced Computing Center, located in Austin, TX also provides allocations for researchers who want to use HPC systems. Home to the [fastest academic supercomputer in the U.S.](#), TACC is a research unit of UT Austin, and it boasts extremely powerful systems:

### **Frontera**

**Compute Nodes:** 8,368 nodes

**CPU:** Intel Xeon Platinum 8280 (“Cascade Lake”)

**Cores per Node:** 56 cores

**Total CPU Cores:** ~468,000+

**Memory per Node:** 192 GB DDR4

**Interconnect:** Mellanox HDR-100 InfiniBand

**Local Storage:** 480 GB SSD/node

### **Lone Star 6**

- **Compute Nodes:** 560 CPU nodes
- **GPU Nodes:** 16 accelerator nodes

### **CPU Nodes**

- **Processor:** 2× AMD EPYC 7763 (Milan)
- **Cores per Node:** 128 cores
- **Memory:** 256 GB DDR4/node
- **Clock Speed:** 2.45 GHz

### **GPU Nodes**

- **GPUs:** 2× NVIDIA A100 (40 GB HBM2) per node

### **Stampede3**

### **Other Texan HPC resources**

Like UT Austin, Texas A&M, UT Dallas, and the University of Houston have HPC clusters of varying size and scale. Although these university-specific clusters are typically only available to students and faculty of the university, they are often accessible through joint research grants, so if you are partnered with one of these institutions, it may be worth examining what resources you have access to.

## National Programs

### NSF Access

The NSF ACCESS (Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support) program is a U.S. National Science Foundation initiative that provides academic researchers, educators, and students with coordinated access to advanced high-performance computing (HPC), cloud computing, storage, data resources, and technical expertise across multiple national computing centers.

Rather than relying on a single supercomputer, ACCESS connects a nationwide ecosystem of systems through a unified allocation and user support framework. **ACCESS is supported through your Texas State ID by default, but you must first create an account through your Texas State ID.**

Through ACCESS, you can explore free HPC resources across the U.S. and receive allocations depending on **proposed projects** of different computational complexity ('Explore', 'Discover', 'Accelerate', and 'Maximize'.) **ACCESS is the most comprehensive and useful tool available to researchers for finding HPC resources through the NSF. If you cannot find compute resources through LEAP2 or TACC, it is highly recommended you submit a project through ACCESS for compute allocations on different HPC systems.**

### NWSC

The NCAR-Wyoming Supercomputing Center (NWSC) is a national high-performance computing facility operated by the National Center for Atmospheric Research (NCAR) in partnership with the National Science Foundation and the State of Wyoming. The NWSC provides advanced supercomputing, data storage, and visualization resources designed primarily to support atmospheric science, climate modeling, Earth system science, and related computational research. Rather than serving a single university, the NWSC operates as an open scientific user facility supporting researchers from universities, government laboratories, and collaborating institutions worldwide whose work aligns with geoscience and environmental research missions.

Access to NWSC computing resources is typically available to academic researchers, students, and research collaborators through NCAR-supported allocation programs. Researchers generally obtain access by submitting a computational project proposal describing their scientific objectives, required resources, and expected impact through NCAR's allocation request process or through NSF-supported programs such as ACCESS.

### Open Science Grid Consortium

The Open Science Grid (OSG) is a nationally distributed high-throughput computing infrastructure operated by the Open Science Grid Consortium and supported by the National Science Foundation and the U.S. Department of Energy. Unlike traditional supercomputers that run tightly coupled simulations on a single cluster, OSG connects computing resources contributed by universities, national laboratories, and research institutions across the United States into a shared federation.

They are partnered with and can be accessed through ACCESS.

## National Labs

Many Department of Energy (DoE) labs offer free computing resources for researchers who submit proposals through their websites. Although access can be competitive, both academic and industry researchers can apply for accounts.

## DOE INCITE

<https://doeleadershipcomputing.org/>

**Key DOE HPC user facilities include, from most to least accessible:**

### *Open to Researchers*

#### Oak Ridge National Labs (OLCF)

**Oak Ridge Leadership Computing Facility**, operated at Oak Ridge National Laboratory, specializes in **capability computing**, where extremely large *simulations* are created using tens of thousands of nodes simultaneously. OLCF is one of the Department of Energy's flagship leadership-class supercomputing centers dedicated to open scientific research. OLCF hosts **Frontier**, the world's first operational exascale supercomputer capable of performing over one quintillion calculations per second, and is often lauded as one of the best supercomputing facilities in the United States.

#### Argonne (ALCF)

The Argonne Leadership Computing Facility (ALCF) provides leadership-scale computing resources for open science at Argonne National Laboratory. Its flagship system, **Aurora**, is one of the world's first exascale supercomputers and supports massive simulations, AI training, and data-intensive scientific workflows. It is **available worldwide, and specializes in HPC + AI integration**.

### *Open to Researchers collaborating with the facility*

#### Lawrence Livermore National Laboratory

Lawrence Livermore National Laboratory operates some of the most powerful simulation systems in the world, primarily supporting national security and advanced physics research. Its newest system, **El Capitan**, became **the world's fastest supercomputer** in 2024–2025 with exascale performance exceeding 1.8 exaflops.

- Although each DoE lab has incredible systems, LLNL is often hailed as the most advanced.

#### Lawrence-Berkely (NERSC)

The National Energy Research Scientific Computing Center (NERSC) is the primary production computing facility for the DOE Office of Science and supports over 11,000 researchers annually across universities and laboratories worldwide. Unlike leadership facilities focused on a few massive jobs, NERSC provides large-scale **throughput computing** for thousands of scientific

projects simultaneously. It is designed for **daily scientific computing rather than large-scale simulations**.

### Los Alamos National Laboratory

Los Alamos National Laboratory operates advanced HPC systems supporting national security, physics, energy, and computational science research. LANL has historically pioneered large-scale parallel computing and continues developing next-generation AI and simulation systems within DOE's exascale ecosystem.

### Sandia National Laboratories

Sandia National Laboratories develops advanced computing platforms supporting engineering, cybersecurity, and national security science. Sandia also hosts experimental computing platforms such as the **QSCOUT quantum computing testbed**, allowing researchers to explore emerging quantum-HPC workflows. It is an excellent choice for **researchers interested in the HPC or quantum computing spaces**.

### National Energy Technology Laboratory

The National Energy Technology Laboratory supports computational research focused on energy systems, carbon capture, advanced manufacturing, and industrial simulation. However, it is worth noting that NETL operates HPC systems tailored toward applied energy research rather than leadership-scale capability computing.

## University/Consortium

Many universities own HPC clusters that can rival resources provided by dedicated HPC facilities. Although access to these can be difficult to obtain unless partnered with these universities, the following university-level resources are particularly notable:

**Massachusetts green high performance computing center** - <https://www.mghpcc.org/>

- For those who are looking to use green computing resources.

The **Massachusetts Green High Performance Computing Center (MGHPCC)** is an intercollegiate high-performance computing facility located in Holyoke, Massachusetts that provides large-scale computational infrastructure powered primarily by energy-efficient and renewable energy sources. Operated as a collaboration between major research universities (Boston University, Harvard, MIT, Northeastern University, the University of Massachusetts system, and Yale), a defining feature of MGHPCC is its focus on **sustainable or “green” computing**, using hydroelectric power and energy-efficient data center design to significantly reduce the environmental footprint of large-scale computation.

Access to MGHPCC resources is typically obtained through membership institutions, collaborative research partnerships, or shared platforms such as the Unity Research Computing Platform, which allows both academic and industry researchers to run scalable workflows using CPUs and GPUs through managed allocations.

**University of Arizona Cluster** - <https://hpcdocs.hpc.arizona.edu/>

**University of Minnesota** - <https://msi.umn.edu/>

**University of Maryland High-Performance computing center** - <https://hpcc.umd.edu/>

Other universities like Georgia Tech, Columbia, Dartmouth, etc. also have supercomputing clusters. If you are working with researchers from a different university, it might be worth looking into what HPC resources they have available at their home institution.

## International

For those who are conducting international research, the following supercomputing centers are particularly noteworthy across Asia and Europe.

**Swiss National Supercomputing Center (CSCS)**

<https://www.cscs.ch/>

**Edinburgh Parallel Computing Center (EPCC)**

<https://www.epcc.ed.ac.uk/>

## Other Platforms

**Community and Academic Distributed Computing Grids (NIH/NASA)**

Community and academic distributed computing grids supported by agencies such as the National Institutes of Health (NIH) and NASA provide shared computational infrastructure designed to support large scientific collaborations and data-intensive research communities.

## Chain of Escalation – How much Compute do I Need?

**Generally, most HPC research follow the sliding scale below from least to most advanced depending on research needs and governmental support:**

University Cluster (LEAP2 / campus HPC)



NSF ACCESS Allocation



TACC / PSC / SDSC systems



NERSC Production Computing



INCITE Award (OLCF / ALCF)



National Lab Collaboration



Leadership / Exascale Science



## Paid Alternatives

Companies like Lambda labs, Coreweave, and others offer HPC/GPU cluster access via contract or subscription (often used for AI/ML workloads). These aren't traditional academic "allocations" but researchers with grant funding can budget for them if appropriate for research tasks.